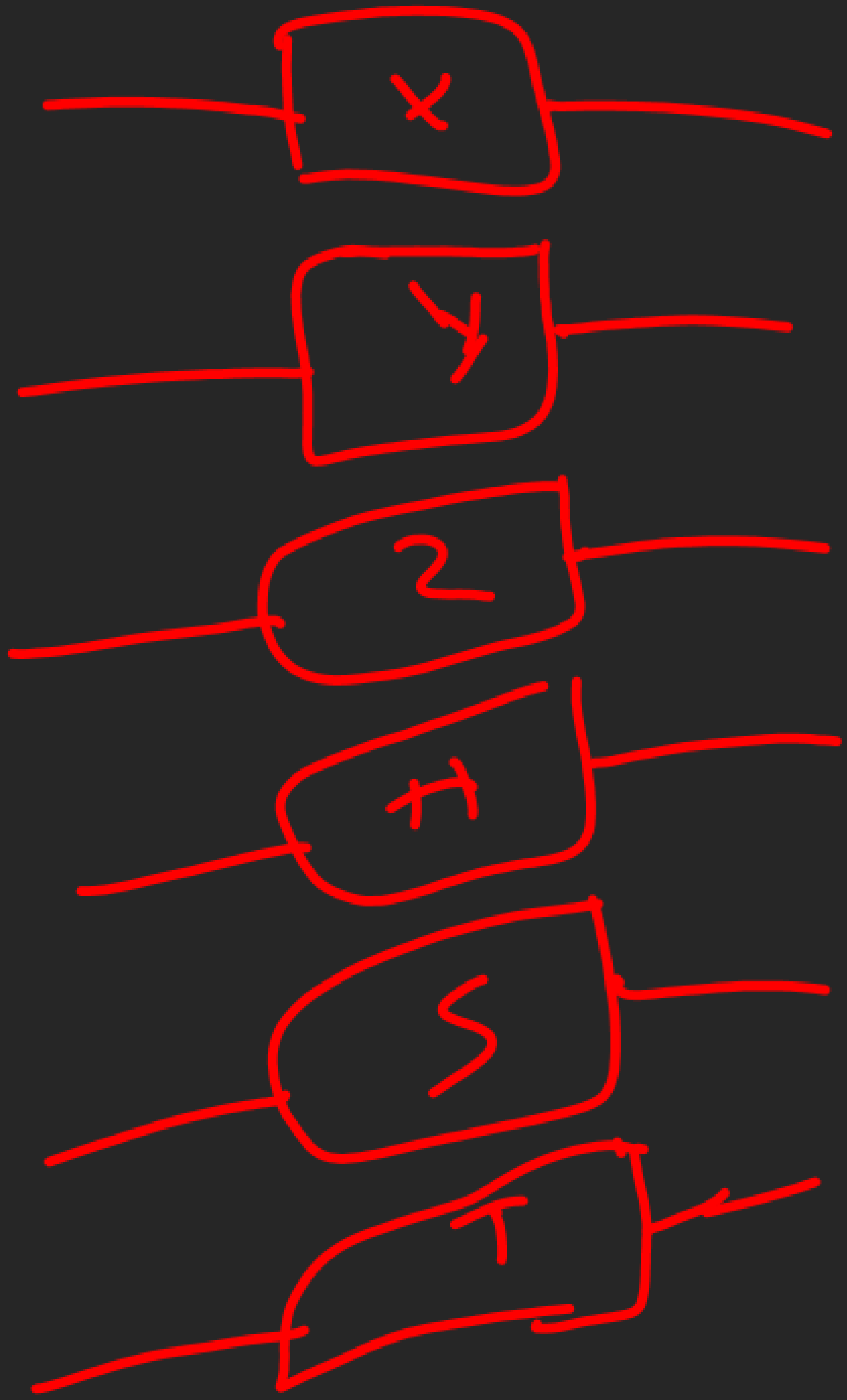
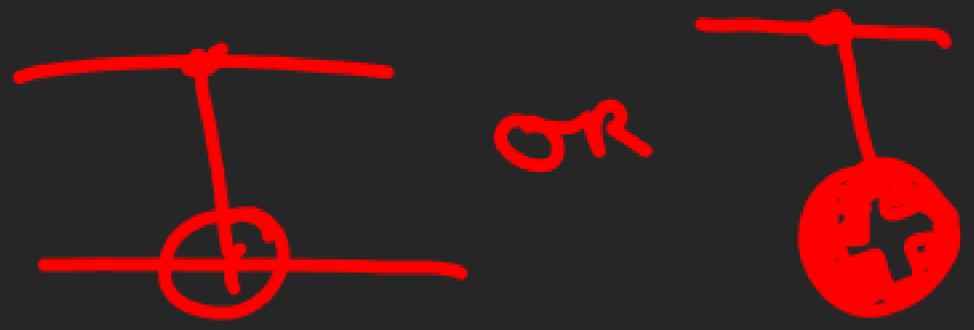
Quantum Circuits

Single Qubit Gates



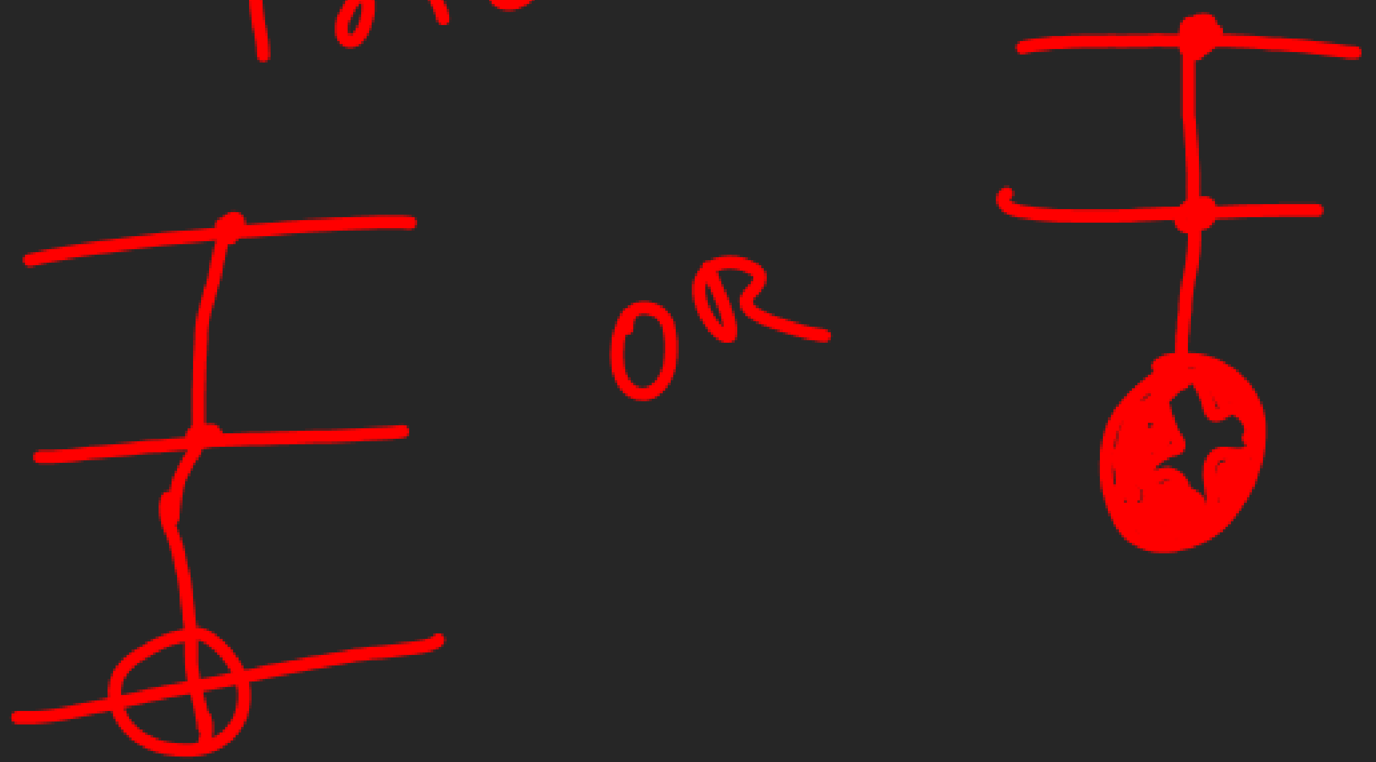
→ Memorize eigenvectors for all of them

CNOT Gate

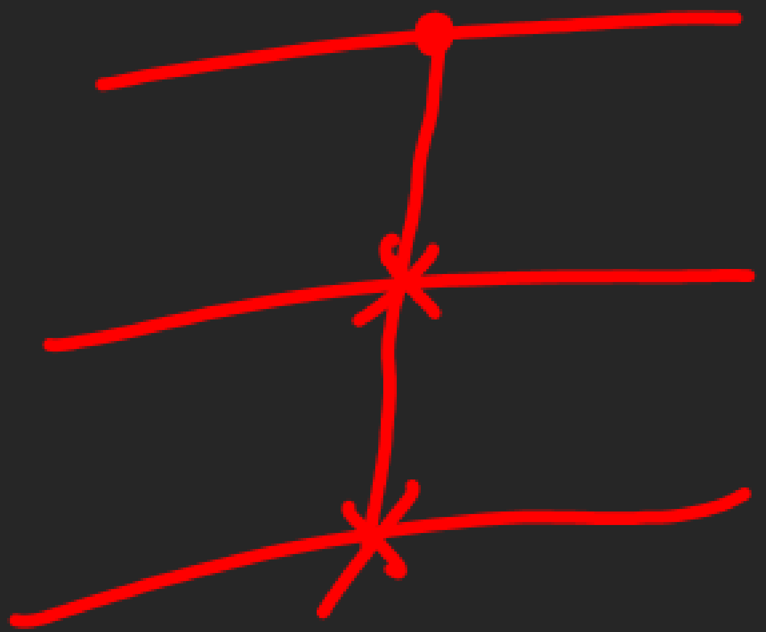


SWAP Gate

Tofolli Gate



Fredkin Gate



$$Y|\psi\rangle = \lambda|\psi\rangle$$

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}\begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \lambda\alpha \\ \lambda\beta \end{bmatrix}$$

$$-i\beta = \lambda\alpha \qquad \alpha^2+\beta^2 = 1$$

$$i\alpha = \lambda\beta$$

$$(-i\beta)^2 = \left(\lambda\sqrt{1-\beta^2}\right)^2$$

$$-\beta^2 = \lambda^2(1-\beta^2)$$

$$\beta^2 = \frac{\lambda^2}{\lambda^2-1} \quad 1$$

$$\left(i\sqrt{1-\beta^2}\right)^2 = (\lambda\beta)^2$$

$$\beta^2 - 1 = \lambda^2\beta^2$$

$$\lambda^2 = \frac{\beta^2-1}{\beta^2} \quad 2$$

$$= 1 - \frac{1}{\beta^2} = 1 - \left(\frac{\lambda^2-1}{\lambda^2}\right)$$

$$= 1 - \left(1 - \frac{1}{\lambda^2}\right)$$

$$= \frac{1}{\lambda^2}$$

$x^4 = 1$

$\downarrow$

Solve Further

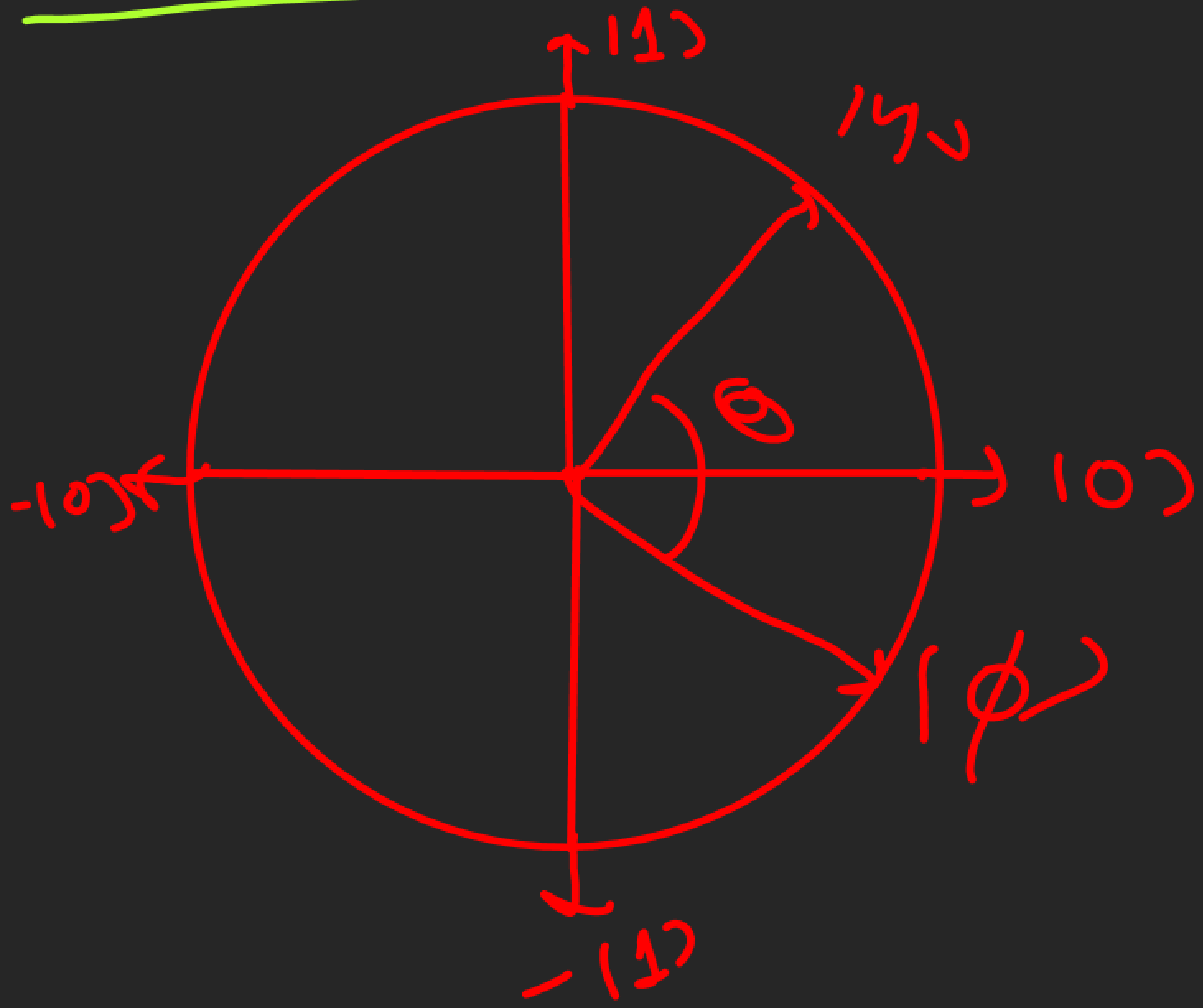
$$|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

$$|-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

$$Y|+i\rangle = +|+i\rangle$$

$$Y|-i\rangle = -|-i\rangle$$

Inner Product



$$\langle\psi|\phi\rangle = \langle\phi|\psi\rangle$$

$$|\langle\psi|\phi\rangle| = |\overline{\langle\phi|\psi\rangle}| = \cos\theta$$

$$\langle\psi|\psi\rangle = \||\psi\rangle\|^2$$

$$\bar{\alpha} = \alpha^*$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\langle\psi| = \bar{\alpha}\langle 0| + \bar{\beta}\langle 1|$$

Cauchy-Schwartz Inequality

$$|\langle\psi|\phi\rangle| \le \| |\psi\rangle \| \cdot \| |\phi\rangle \|$$

1. $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$
   $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle$

2. $|\psi\rangle = |\psi_\alpha\rangle$
   $|\phi\rangle = |\psi_\beta\rangle$

$$|\bar{\alpha}\gamma + \bar{\beta}\delta| \le \sqrt{\alpha^2+\beta^2} \cdot \sqrt{\gamma^2+\delta^2}$$

$$|\bar{\alpha}\gamma + \bar{\beta}\delta| \le 1$$

$\alpha = \cos\alpha$ $\quad$ $\gamma = \cos\beta$

$\beta = \sin\alpha$ $\quad$ $\delta = \sin\beta$

$$\Rightarrow |\bar{\alpha}\gamma + \bar{\beta}\delta| \le 1$$

$$|\overline{\cos\alpha}\cos\beta + \overline{\sin\alpha}\cdot\sin\beta| \le 1$$

Equality holds if and only if $|\psi\rangle$ and $|\phi\rangle$ are linearly dependent

$a|\psi\rangle + b|\phi\rangle = 0 \rightarrow$ Linearly dependent

$\downarrow$

Parellel or Anti-Parellel

$\langle\psi|\phi\rangle = 0 \rightarrow |\psi\rangle$ and $|\phi\rangle$ are orthogonal

$\langle A, B\rangle = Tr(A^{\dagger}B)$
Hilbert-Schmidt Inner Product

$\{|\psi_1\rangle, \_\_\_ |\psi_n\rangle\}$

$\downarrow$

orthonormal Basis

IF $\langle\psi_i|\psi_j\rangle = \delta_{ij}$ $\quad i,j \in \{1, \_\_ n\}$

Ex. $\rightarrow \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$\rightarrow \{|\phi^+\rangle, |\phi^-\rangle, |\psi^+\rangle, |\psi^-\rangle\}$

Rows and Columns of a Unitary matrix are orthonormal sets

Theorem

$\{|\psi_1\rangle \_\_\_\_ |\psi_m\rangle\} \rightarrow$ Orthonormal Basis

for $m < n$ $\{|\psi_{m+1}\rangle \_\_\_\_ |\psi_n\rangle\}$ can be formed

such that $\{|\psi_1\rangle \_\_\_\_ |\psi_n\rangle\}$ is an orthonormal basis

Read Gram-Schmidt Orthogonalisation process

Projection Matrix

Conditions $\rightarrow P = P^{\dagger}$

$\rightarrow P^2 = P$

$|\psi\rangle \rightarrow$ unit vector $\qquad |\psi\rangle\langle\psi| \rightarrow$ Projection matrix

$$P = \sum_{k=1}^{r} |\psi_k\rangle\langle\psi_k|$$

for some orthonormal basis $\{|\psi_1\rangle \text{ — } |\psi_n\rangle\}$

$$\{P_1 \text{ — } P_n\} \rightarrow \sum_{k=1}^{n} P_k = I$$

$\downarrow$

projective measurement

$Pr(\text{outcome } m) = p(m) = \| P_m |\psi\rangle \|^2 = \langle\psi| P_m |\psi\rangle$

Posterior State → $\dfrac{P_m|\psi\rangle}{\| P_m|\psi\rangle \|}$

$P_m = |m\rangle\langle m| \otimes I$ → For multi-qubit system

$H = \sum_\lambda \lambda P_\lambda$

λ → Eigenvalues

$|\lambda\rangle$ → Eigenvectors

$P_\lambda = |\lambda\rangle\langle\lambda|$

Any Hermitian matrix

↳ Spectral decomposition

Global Phase

$|\phi\rangle = e^{i\theta}|\psi\rangle$

Probablities of $|\phi\rangle$ and $|\psi\rangle$ are are the same

$e^{i\theta} \rightarrow$ Global Phase

$\downarrow$

we ignore it

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$|\phi\rangle = e^{i\theta}\alpha|0\rangle + e^{i\theta}\beta|1\rangle$

$|\psi\rangle \rightarrow P(0) = |\alpha|^2$

$P(1) = |\beta|^2$

$|\phi\rangle \rightarrow P(0) = |e^{i\theta}\cdot\alpha|^2 = |\alpha|^2$

$P(1) = |e^{i\theta}\cdot\beta|^2 = |\beta|^2$

No Cloning Theorem

$$\forall |\psi\rangle = U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$U \to$ doesn't exist

Perfectly Discriminate States

If two quantum state $|\psi\rangle$ and $|\phi\rangle$ a . perfectly discriminate, then they must be orthonormal

$$\langle\psi|\phi\rangle = 0$$

perfectly discriminate $\rightarrow$ If $U|\psi\rangle = 0$ then $U|\phi\rangle = 1$

$$U = \begin{pmatrix} | & | & \vdots & \vdots & | \\ |\psi\rangle & |\phi\rangle & | & | & | \\ | & | & | & | & | \end{pmatrix}$$

Any elements such that U is unitary